Report R15 — Can an H&E Slide Give You the Methylome?

760 matched TCGA lung patients. Partly yes, modestly, and not as a substitute for the assay.

HELD FOR IP — not for publication pending patent filing.

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Summary

The companion-diagnostic product requires a slide to go in and a molecular report to come out. Everything the programme had built did the two halves separately: biomarkers from slides (R12/R13), methylation from arrays (R08/R10). This report tests the join, on 760 TCGA patients who have both Phikon-v2 whole-slide features and 450k methylation — 413 adenocarcinoma, 347 squamous, across 67 tissue-source sites.

H&E does carry epigenomic information, and it is not merely histological subtype. Six fixed, annotation-defined methylation summaries were each predicted from morphology under site-grouped cross-validation, and all six retain signal after controlling for LUAD-versus-LUSC status (partial Spearman ρ 0.135–0.244, every one significant, strongest $p = 9 \times 10^{-12}$).

But it does not replace the assay. KEAP1 mutation status reaches AUROC 0.664 (95% CI 0.604–0.722) from the slide, against **0.910** from the methylation array in R10 — a shortfall of 0.246. The correlations above, at $\rho \approx 0.2$–0.3, are real associations, not measurement-grade prediction.

A methodological result worth more than either. The first attempt at this question returned a clean negative. We supervised the slide embedding on subtype and then asked it about methylation; it learned nothing useful for the second task and was outperformed by the subtype label alone — a 512-dimensional embedding beaten by one binary variable. Training the same architecture directly on methylation reversed the answer *on the six aggregate targets.*

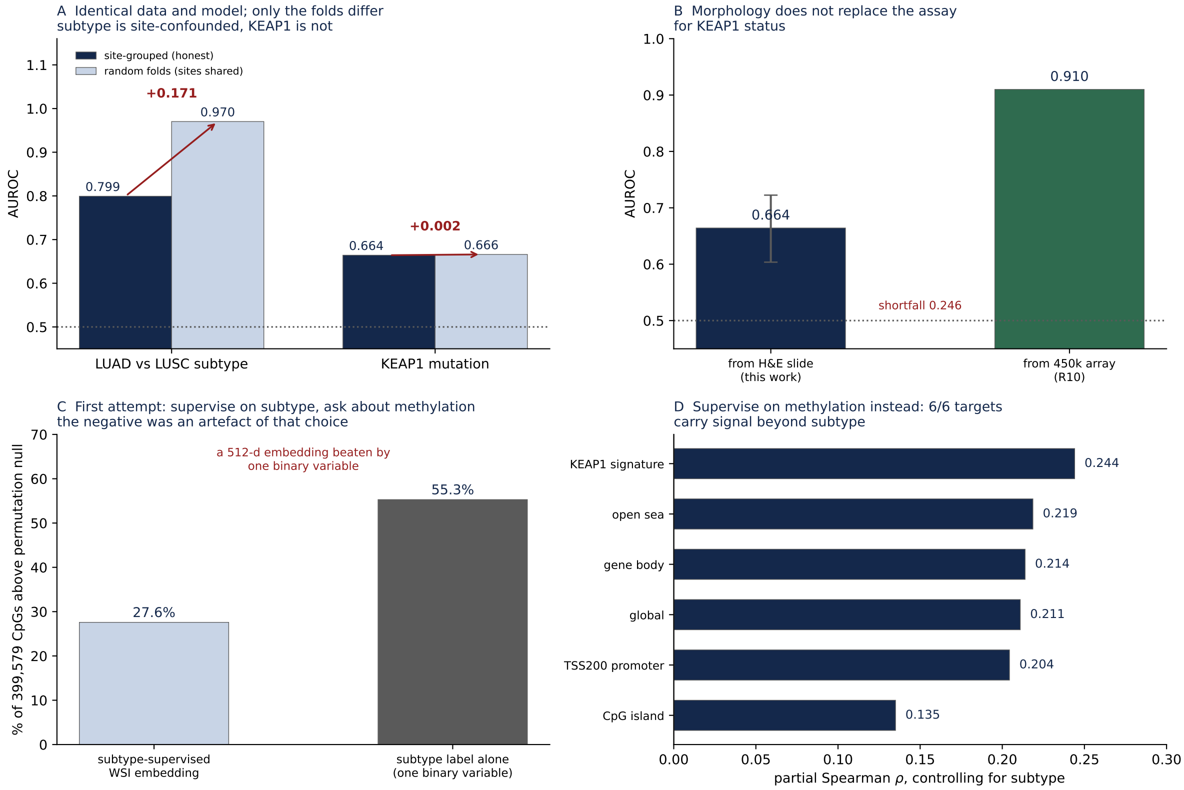
Corrected 2026-08-17 (R17). This paragraph originally read "*the negative was an artefact of our own design choice*" without qualification, and §5 below claimed the reversal generally. R17 re-ran the genome-wide scan on the methylation-supervised embedding and it does **not** reverse: 83,369 CpGs above null against 90,137 for the subtype-supervised embedding, median ρ 0.0073 against 0.0057. The reversal is specific to the aggregate targets. R17 also shows why — a patient's *observed* global mean methylation alone puts 74.7% of CpGs above null, so five of our six targets being means over large annotation classes is enough for a representation to learn that one axis while carrying nothing at CpG resolution.

1 The site-leakage measurement

Before any biological claim: this cohort let us isolate how much apparent performance comes from train and test sharing contributing institutions. Data, architecture, hyperparameters and seeds were held identical and only the cross-validation partition changed.

	Site-grouped	Random folds	Inflation
LUAD vs LUSC subtype (AUROC)	0.7992	0.9703	**+0.1710**
KEAP1 mutation (AUROC)	0.6642	0.6660	+0.0018
methylation targets (Spearman ρ)			
KEAP1 signature	0.3172	0.3927	+0.0755
global mean	0.2897	0.3680	+0.0783
CpG island	0.2182	0.3358	+0.1176
open sea	0.2868	0.3478	+0.0610
TSS200 promoter	0.3029	0.4212	+0.1184
gene body	0.2988	0.3577	+0.0589
mean methylation inflation			**+0.0849**

Three things follow. The random-fold subtype figure, 0.9703, reproduces an independent centralized benchmark of 0.970 to within 0.0003, which establishes that the site-grouped numbers are honest out-of-site performance and not a weak pipeline. Methylation is site-confounded, as batch effects would predict, but less so than subtype: a naive analysis would have reported $\rho \approx 0.34$–0.42 where the honest values are 0.22–0.32. And KEAP1 mutation is not site-confounded at all, because mutation status — unlike subtype prevalence — has no institutional structure to exploit.



A Fold assignment alone, on identical data and model. **B** KEAP1 from slide against KEAP1 from array. **C** The first, negative answer: a subtype-supervised embedding loses to the subtype label. **D** Supervising on methylation instead — partial correlations after removing subtype.

2 Targets, and why they are defined this way

Six summaries, each a mean β over a *fixed, annotation-defined* probe set. Nothing was fitted to the data, so there is no selection to argue about:

Target	Probes	Mean β	Definition
KEAP1 signature	115,517	0.519	R10's KEAP1-differential probe list
global	399,579	0.473	all retained probes
CpG island	134,095	0.226	manifest island annotation
open sea	135,384	0.668	manifest open-sea annotation
TSS200 promoter	47,068	0.172	manifest gene-region annotation
gene body	133,165	0.628	manifest gene-region annotation

The means sanity-check against known biology — promoters and islands hypomethylated, open sea and gene bodies hypermethylated — which is a cheap check that the matrix is assembled correctly.

3 Does morphology add beyond histological subtype?

Subtype is separable from methylation at AUROC 0.955 (R08), so "H&E predicts methylation" could trivially mean "H&E predicts subtype, and subtype predicts methylation." A pathologist supplies subtype for free, so the question that matters commercially is whether morphology adds anything on top of it.

Target	ρ (WSI)	ρ (subtype)	Partial ρ	p
KEAP1 signature	0.317	0.257	**0.244**	9.1×10^{-12}
open sea	0.287	0.219	**0.219**	1.1×10^{-9}
gene body	0.299	0.270	**0.214**	2.6×10^{-9}
global	0.290	0.264	**0.211**	4.3×10^{-9}
TSS200 promoter	0.303	0.345	**0.204**	1.3×10^{-8}
CpG island	0.218	0.288	**0.135**	1.9×10^{-4}

All six. Note that on two targets — promoter and island — the subtype label alone out-predicts the slide, yet the partial correlation remains positive: morphology carries *different* information rather than more of the same.

4 The first attempt, and why it was wrong

Recorded because the sequence is the most transferable thing here.

The first run supervised the attention-MIL on subtype, extracted the 512-dimensional attention-pooled embedding out-of-fold, then regressed methylation on it. Results: genome-wide median $\rho = 0.006$; 110,212 of 399,579 CpGs above the permutation null (27.6%) against **220,925 (55.3%)** for the subtype label by itself; bulk methylation $\rho = 0.011$, $p = 0.76$; and genomic-context enrichment flat at 0.97–1.01 across island, shore, shelf and open sea, where R10's KEAP1 phenotype had been strongly structured (island 0.54, open sea 1.44).

Every one of those readings pointed the same way, and the conclusion — morphology does not carry methylation *at all* — was too broad. The embedding had been optimised to separate two histological classes, and nothing else was asked of it during training. Supervising on methylation directly recovered signal *on the six aggregate targets.*

An honest negative therefore needs its supervision examined before it is believed.

Corrected 2026-08-17 (R17). Two sentences here originally read "supervising on methylation directly recovered the signal" and "that flatness of genomic context . . . was in fact evidence that the representation contained no methylation-relevant structure to find". Both are withdrawn. R17 ran this exact scan on the methylation-supervised embedding: it is not better (83,369 above null against 90,137), and its genomic context is still nearly flat — spread 0.212 against 0.047 — and tilted the *opposite* way to R10, with islands enriched at 1.116 where R10 had them depleted at 0.54. So flatness here was not a diagnostic of the wrong supervision. The genome-wide negative is robust to supervision; see `results/R17-percpg-methylation-supervised/`.

5 Commercial reading

- **A slide cannot replace a methylation assay.** KEAP1 at 0.664 from morphology against 0.910 from the array. The product is slide *plus* assay, not slide instead of assay.

- **Morphology is genuinely informative about the epigenome**, at $\rho \approx 0.2$ beyond subtype. That is a real mechanistic finding and is the patentable content here, but it is an association, not a measurement.

- **This inflation is encoder-specific. Added 2026-08-17 (R18).** The +0.0849 below was re-measured on `facebook/dinov2-large` with identical tiles, folds, architecture and hyperparameters: it becomes **+0.0004**, with three of the six targets moving in the anti-leakage direction. On fraction of remaining headroom, 0.119 for Phikon-v2 against 0.000. This is not a headroom artefact — the same comparison shows *subtype* leakage reproducing and growing (+0.2034 against +0.1710). So subtype leakage is a property of the archive and *methylation* leakage was a property of Phikon-v2 features. Every methylation-inflation figure in this report should be read as specific to this encoder. See `results/R18-encoder-robustness/`.

- **Any naive validation of this capability would have overstated it** by about 0.085 in ρ, and by 0.171 in AUROC for subtype. A vendor reporting single-collection cross-validated numbers is reporting an upper bound.

6 Tumour purity: the most likely benign explanation, tested

The first version of this report named tumour purity, stromal fraction and proliferation as the most plausible innocent driver of the association — all three plausibly move both morphology and bulk methylation — and stated that no correction was available. That was premature. TCGA publishes consensus ABSOLUTE purity estimates, available here for **741 of 760 patients** (median purity 0.460).

The confound is real. Purity is strongly associated with observed methylation ($\rho = -0.306$, $p = 1.5 \times 10^{-17}$ for the KEAP1 signature) and the morphology-derived predictions do track it ($\rho = -0.218$, $p = 2.2 \times 10^{-9}$). So the concern was not hypothetical.

Adjusting for it barely moves the result.

Target	**Partial ρ** (subtype only)	**+ purity** (subtype + purity)	**Change**
keap1_sig	0.252	0.221	-0.031
global	0.218	0.210	-0.008
island	0.134	0.136	+0.002
opensea	0.228	0.218	-0.010
tss200	0.210	0.190	-0.021
body	0.220	0.221	+0.001

The largest change is the KEAP1 signature, from 0.252 to 0.221. Every target remains significant (all $p \leq 2.1 \times 10^{-4}$; four at $p < 10^{-8}$). So morphology carries information about methylation that is not explained by histological subtype and not explained by how much tumour is on the slide.

This does not exhaust the confounders — stromal composition, necrosis and proliferation index are still unadjusted — but it removes the one with the strongest prior and the best available measurement.

7 Limitations

Association, not mechanism. Nothing here shows morphology is caused by methylation state or vice versa. Tumour purity is now adjusted for (§6) and the association survives; stromal composition, necrosis and proliferation remain unadjusted.

Modest magnitudes. Partial ρ 0.135–0.244 explains a few percent of variance. Presented as evidence of information, not of utility.

Subtype-supervised embeddings for the first arm only. The negative in §5 is bounded to that design and should not be quoted as a general result about H&E.

One cohort, one encoder, one array platform. TCGA lung, Phikon-v2, Illumina 450k. Nothing replicated. The programme has learned (R09) not to assume a replication cohort exists.

Six targets, not the full methylome. Fixed annotation-defined summaries were chosen for interpretability and leakage-safety. Per-CpG prediction under methylation supervision was not attempted here — it was attempted in R17 (2026-08-17) and is negative, which bounds every claim in this report to the aggregate targets.

Mean-pooled slide aggregation per patient. Patients with multiple slides are averaged; no attempt was made to weight by tumour content.

8 Reproducing this report

- Analysis: `evidence/m13_wsi_to_methylation.py` (arms and gates); `evidence/mil_embed.py` (subtype-supervised MIL); `evidence/mil_meth.py` and `mil_meth_random.py` (methylation-supervised, site-grouped and random arms)
- Config hash, written before any outcome was read: `3d8b9658f679e069...2c5a5ef3`
- `evidence/meth_leakage_arm.json` — the grouped/random comparison for all six targets
- `evidence/per_cpg_rho_{grouped,null,subtype_only}.npy` — all 399,579 per-CpG correlations from the first arm
- Figure: `python3 make_figure.py`